

Software Bill of Materials (SBOM)

This document outlines the core firmware dependencies, third-party libraries, and toolchains required to build the Magnetic Field Scanner v5.x.x.

Core Toolchain & Framework

Component	Version / Target	Description
Platform Framework	Arduino ESP32 Core v3.x.x	Built upon ESP-IDF v5.1. Provides FreeRTOS and base hardware abstractions.
Hardware Target	ESP32-S3 Dev Module	16MB Flash, OPI PSRAM, QIO 80MHz, Hardware CDC + JTAG.

Third-Party Libraries

Library Name	Version	License	Description / Purpose
LVGL (Light and Versatile Graphics Library)	v8.3.x	MIT	Drives the core GUI, arc gauges, dynamic color mapping, and capacitive touch handling. Must be configured via <code>lv_conf.h</code> .

Library Name	Version	License	Description / Purpose
ArduinoJson	v7.0+	MIT	Used for parsing and serializing <code>settings.json</code> and <code>calibration.json</code> to/from the SD Card and FFat filesystems.
Madgwick AHRS	v1.2.0 (or embedded)	GPL	Sensor fusion algorithm that combines Accelerometer and Gyroscope data into reliable Euler angles (Pitch/Roll/Yaw) to drive the radar projection map.
TCA9554	<i>Custom/Embedded</i>	N/A	I2C I/O Expander driver (Address 0x20) used to control LCD backlight, display reset, and PMIC <code>SYS_EN</code> latch.
ES8311 / ES7210	<i>Custom/Embedded</i>	N/A	I2S Audio Codec drivers utilized by the Waveshare board for dynamic frequency audio feedback.

Library Name	Version	License	Description / Purpose
RM3100 Driver	Custom/Embedded	N/A	Heavily modified to operate exclusively in POLL mode across the I2C bus (0x21, 0x22, 0x23) with raw integer-to-nT domain shifting.

Auxiliary Python Scripts

Found in the `/scripts` directory for system characterization and calibration analysis.

Package Name	Version	Purpose
Python	<code>>= 3.9</code>	Base execution environment.
pandas	latest	High-performance CSV ingestion and DataFrame manipulation for log analysis.
numpy	latest	Matrix mathematics and Least-Squares Kabsch ellipsoid fitting.
matplotlib	latest	Generation of 2D/3D plots and calibration visualizations.

Package Name	Version	Purpose
scipy	latest	Advanced statistical metrics (variance, standard deviation) for hardware limit modeling.